Dipankar Sarma, distinguished engineer, Systems & Technology Labs, IBM India

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Q:Can you explain the function of the Linux Development Centre in the Indian and global context?

The Linux Technology Centre, as I mentioned, is a core technology development organisation, which is a part of IBM's Systems & Technology Group. Our main function is to look at some of these key open source projects and make them run on our platforms. We work on the Linux kernel predominantly and we team up with our distribution partners like Red Hat and SUSE to make Linux better, so that those distros run much better on our platforms. We contribute the newer technologies that we develop to the open source community and we benefit from the fact that due to our efforts and contributions, Linux as a distribution works considerably better on our platforms.

Even though Linux is the dominant project in the world of open source technology, it is not the only project that we contribute to. Kernel-based Virtual Machine (KVM), which is about virtualisation using the Linux system, is another project that we contribute to very significantly. It is used on IBM's platforms extensively. A lot of our customers make use of KVM as well. Samba is also a project to which IBM has made a lot of contributions. In recent times, it is OpenStack that has become a very important project from our perspective, and there are a number of people within IBM who are contributing to it.

We have a strong team of engineers at this IBM centre in India. We have a history of about 12 years. The centre was started in 2000 and from then onwards, we have made significant contributions to the Linux kernel from India itself. In fact, when we started, there weren't many open source contributors from India, especially not in the Linux kernel domain. My team was probably one of the earliest to start contributing to the Linux kernel from India. Over the years, that team has grown significantly and the contributions that we have made have also grown very significantly. There are a number of key contributors to the Linux kernel who are based in our lab in India and they are fairly well known in the Linux kernel community. If you look at our track record, it is reasonably easy to find out where these people are based. You can see that a fairly large number of contributions like Read-Copy Update locking, Group-aware scheduler, Cgroups, CPU hotplug, Power-aware scheduler, etc, are done by the LTC India team based out of our labs.

Q:At IBM, do you get enough open source talent?

India has a very good history of using and contributing to open source technology. It may have started about 10-12 years ago but, over the years, it has picked up extremely well. There is a study done by the United Nations University in 2006, which showed India as a dominant contributor to the open source projects. PC and computer penetration have a significant role in open source contribution. When you look at the contributions from India by per capita income of the country, you will find that India is a dominant contributor. Also, there is a lot of focus on embedded systems and mobile phone technology development in India, resulting in the development of a lot of local talent.

As far as the industry ecosystem is concerned, and within academia as well, I think we have seen a significant shift towards using open source technology in different ways. There are a number of colleges where we have seen work on open infrastructures. We have seen colleges adopting OpenStack as their internal cloud implementation, to be used for research as well as for convenience within their organisations. We are also seeing Linux being increasingly used in the classroom. Many of the top institutes like the IITs and NITs use Linux for research work as well as for class assignments. They include Linux for the basic C programming classes as well as for complex academic courses, including operating systems and computer architectures. We see a lot of that happening in various universities in India and, hence, we have been able to get people who come to us already trained in many skills in the area of Linux and open source. We've seen these really positive signs in the Indian open source ecosystem.